

# **Basic Networking Skills for Digital Forensics**

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**2025/FWSD/11475**

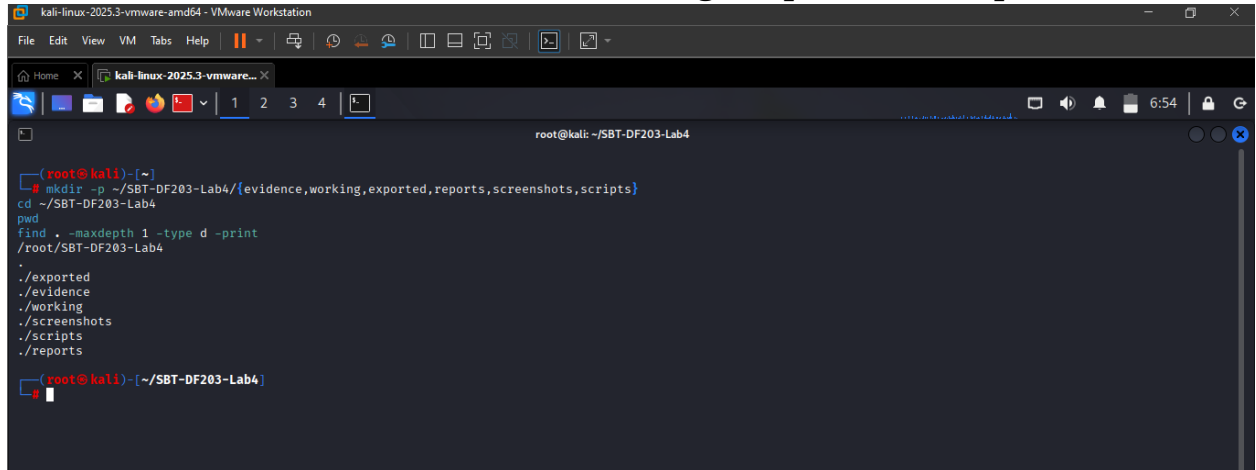
**Fellowship in Web Application Security & Digital Forensics**

**Aminu Idris**

**14/09/2026**

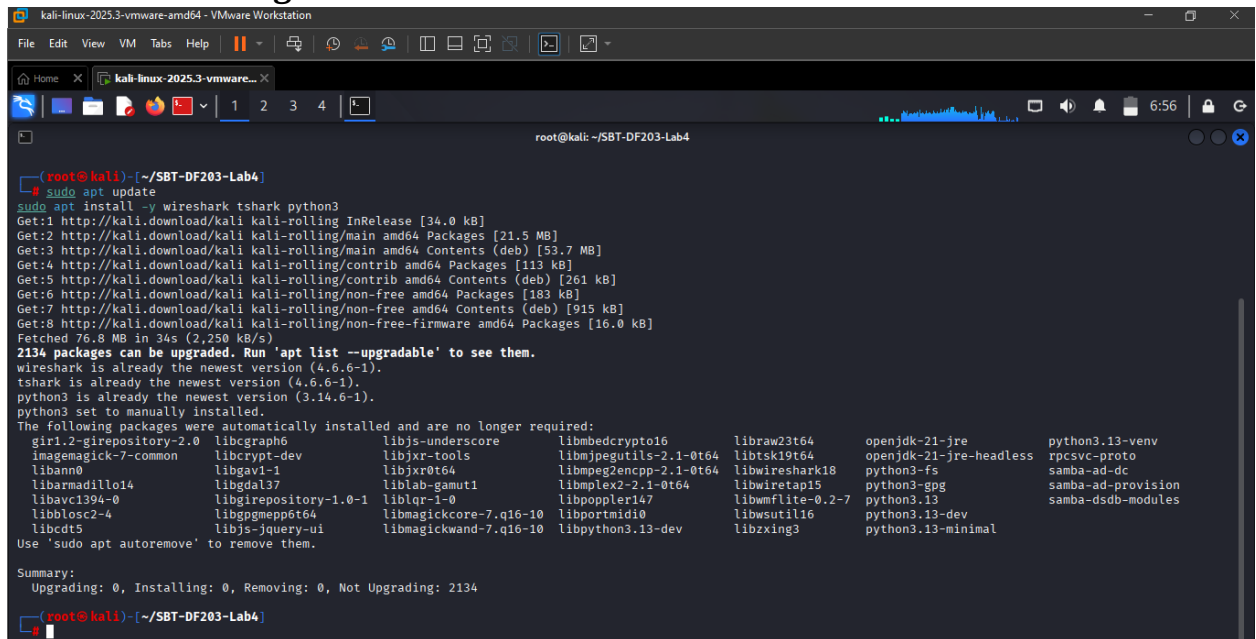
## 1. Lab Folder Structure

Lab folder structure has been initiated using the provided script



```
(root@kali)-[~]
└─$ mkdir -p ~/SBT-DF203-Lab4/{evidence,working,exported,reports,screenshots,scripts}
cd ~/SBT-DF203-Lab4
pwd
/root/SBT-DF203-Lab4
find . -maxdepth 1 -type d -print
./exported
./evidence
./working
./screenshots
./scripts
./reports
(root@kali)-[~/SBT-DF203-Lab4]
```

## 2. Resources installation (wireshark, tshark and python) and captured data downloading



```
(root@kali)-[~/SBT-DF203-Lab4]
└─$ sudo apt update
Get:1 http://kali.download/kali kali-rolling InRelease [34.0 kB]
Get:2 http://kali.download/kali kali-rolling/main amd64 Packages [21.5 MB]
Get:3 http://kali.download/kali kali-rolling/main amd64 Contents (deb) [53.7 MB]
Get:4 http://kali.download/kali kali-rolling/contrib amd64 Packages [113 kB]
Get:5 http://kali.download/kali kali-rolling/contrib amd64 Contents (deb) [261 kB]
Get:6 http://kali.download/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:7 http://kali.download/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Get:8 http://kali.download/kali kali-rolling/non-free-firmware amd64 Packages [16.0 kB]
Fetched 76.8 MB in 34s (2,250 kB/s)
2134 packages can be upgraded. Run 'apt list --upgradable' to see them.
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
python3 is already the newest version (3.14.6-1).
python3 set to manually installed.
The following packages were automatically installed and are no longer required:
gir1.2-girepository-2.0 libcgmath6 libjs-underscore libmbcrypto16 libraw23t64 openjdk-21-jre python3.13-venv
imagemagick-7-common libcrypt-dev libjxr-tools libmjpegutils-2.1-0t64 libtsk19t64 openjdk-21-jre-headless rpcsvc-proto
libann0 libgavi-1 libjxr0t64 libmpeg2encpp-2.1-0t64 libwireshark18 python3-fs samba-ad-dc
libarmadillo14 libgdal37 liblab-gamut1 libmplex2-2.1-0t64 libwiretap15 python3-gpg samba-ad-provision
libavc1394-0 libgirepository-1.0-1 liblqr-1-0 libpoppler147 libwmflite-0.2-7 python3.13 samba-dsdb-modules
libblosc2-4 libgpgmepp6t64 libmagiccore-7.q16-10 libportmidi0 libwsutil16 python3.13-dev
libcdt5 libjs-jquery-ui libmagicwand-7.q16-10 libpython3.13-dev libximg3 python3.13-minimal
Use 'sudo apt autoremove' to remove them.

Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 2134
(root@kali)-[~/SBT-DF203-Lab4]
```

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab4/evidence
root@kali)~/(SBT-DF203-Lab4)
# wget -O evidence/smtp.pcap 'https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap'
--2026-09-14 06:57:43-- https://wiki.wireshark.org/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving wiki.wireshark.org (wiki.wireshark.org) ... 104.26.11.240, 104.26.10.240, 172.67.75.39, ...
Connecting to wiki.wireshark.org (wiki.wireshark.org)|104.26.11.240|:443 ... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap [following]
--2026-09-14 06:57:46-- https://gitlab.com/wireshark/wireshark/-/wikis/uploads/__moin_import__/attachments/SampleCaptures/smtp.pcap
Resolving gitlab.com (gitlab.com) ... 172.65.251.78, 2606:4700:90:0:f22e:fbec:5bed:a9b9
Connecting to gitlab.com (gitlab.com)|172.65.251.78|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 27850 (27K) [application/octet-stream]
Saving to: 'evidence/smtp.pcap'

evidence/smtp.pcap      100%[=====] 27.20K  --KB/s  in 0.002s

2026-09-14 06:57:47 (17.1 MB/s) - 'evidence/smtp.pcap' saved [27850/27850]

root@kali)~/(SBT-DF203-Lab4)
# ls
evidence exported reports screenshots scripts working
root@kali)~/(SBT-DF203-Lab4)
# cd evidence
root@kali)~/(SBT-DF203-Lab4/evidence)
# ls
smtp.pcap
root@kali)~/(SBT-DF203-Lab4/evidence)
```

### 3. Hashes capturing

- a. 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab (evidence/smtp.pcap)
- b. 17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab (working/smtp\_working.pcap)

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab4
root@kali)~/(SBT-DF203-Lab4)
# cp --preserve=timestamps evidence/smtp.pcap working/smtp_working.pcap
root@kali)~/(SBT-DF203-Lab4)
# sha256sum evidence/smtp.pcap working/smtp_working.pcap | tee reports/smtp_capture_hashes.txt
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab evidence/smtp.pcap
17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab working/smtp_working.pcap
root@kali)~/(SBT-DF203-Lab4)
```

The original and working copy matches successfully

### 4. Captured information evidence/smtp.pcap

```

root@kali: ~/SBT-DF203-Lab4
root@kali:~/SBT-DF203-Lab4# capinfos evidence/smt.pcap | tee reports/smt_capinfos.txt
File name:          evidence/smt.pcap
File type:          Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit:  file hdr: 65535 bytes
Number of packets:  60
File size:          27 kB
Data size:          26 kB
Capture duration:   9.198384 seconds
Earliest packet time: 2009-10-05 02:06:07.492060
Latest packet time:  2009-10-05 02:06:16.690444
Data byte rate:     2,920 bytes/s
Data bit rate:      23 kbps
Average packet size: 447.77 bytes
Average packet rate: 6 packets/s
SHA256:             17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a86a47ab
SHA1:               3def1a1fed849e7f23b66e6925dfff05845a400b
Strict time order:  True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 65535
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 60
  
```

## 5. Findings (based on the information above)

1. The capture contains 60 packets.
2. The capture uses Ethernet encapsulation
3. The evidence is a PCAP network captured file
4. The total file size is approximately 27 kB
5. The capture occurred on 5 October 2009.
6. Approximately 26 kB of the file represents captured data.
7. The earliest recorded packet occurred at 02:06:07.492060.
8. And the latest recorded packet occurred at 02:06:16.690444
9. The total capture duration was 9.1 seconds
10. The average packet size was 447.77 bytes.
11. The average packet rate was approximately 6 packets per second
12. The recorded data rate was 2,920 bytes/second
13. The recorded data bit rate was approximately 23 kbps
14. A SHA-256 hash was recorded for evidence integrity verification
15. A SHA-1 hash was also recorded for integrity/reference purposes.

## 6. Mini Evidence and Chain of Custody worksheet

| Field               | My Entry                                 |
|---------------------|------------------------------------------|
| Case/lab identifier | SBT-DF203-Lab3-Idriss-Muhammad-Abdullahi |

|                                    |                                                                                                  |
|------------------------------------|--------------------------------------------------------------------------------------------------|
| <b>Trainee name</b>                | <b>Idriss Muhammad Abdullahi</b>                                                                 |
| <b>Date and time started</b>       | <b>14/09/2026 00.00</b>                                                                          |
| <b>Evidence file name(s)</b>       | <b>smtp.pcap</b>                                                                                 |
| <b>Source or generation method</b> | <b>SMTP</b>                                                                                      |
| <b>Original SHA-256</b>            | <b>17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a<br/>(evidence/smtp.pcap)</b>        |
| <b>Working-copy SHA-256</b>        | <b>17ad230db1b6fd5dd18eb311092df1cf6eb162054bdb47697b89bef5a<br/>(working/smtp_working.pcap)</b> |
| <b>Analysis workstation/V M</b>    | <b>Kali Linux</b>                                                                                |
| <b>Notes on any changes</b>        | <b>No modification that occurs</b>                                                               |

## **Part A - Inventory the Capture and Locate SMTP Streams**

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab4
root@kali) [~/SBT-DF203-Lab4]
# tshark -r working/smtp_working.pcap -q -z conv,tcp | tee reports/tcp_conversations.txt
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214

TCP Conversations
Filter:<No Filter>

10.10.1.4:1470 <-> 74.53.140.153:25
Frames Bytes | Frames Bytes | Frames Bytes | Relative | Duration |
25 1,980 bytes | 28 22 kB | 53 24 kB | 0.036986000 | 7.5777 |

root@kali) [~/SBT-DF203-Lab4]
# tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time -e ip.src -e tcp.srcport -e ip.dst -e tcp.dstport -e _ws.col.info \
| tee reports/smtp_packet_inventory.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
2009-10-05T02:06:08.219663000-0400 74.53.140.153 25 10.10.1.4 1470 S: 220-xc90.websitewelcome.com ESMTP Exim 4.69 #1 Mon, 05 Oct 2009 01:05:54 -0500 | We do not authorize the use of this system to transport unsolicited, and/or bulk e-mail.
2009-10-05T02:06:08.224809000-0400 10.10.1.4 1470 74.53.140.153 25 C: EHLO GP
2009-10-05T02:06:08.566183000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250-xc90.websitewelcome.com Hello GP [122.162.143.157] | SIZE 52428
2009-10-05T02:06:08.568729000-0400 10.10.1.4 1470 74.53.140.153 25 C: AUTH LOGIN
2009-10-05T02:06:08.911081000-0400 74.53.140.153 25 10.10.1.4 1470 S: 334 VXNlcm5hbWU6
2009-10-05T02:06:08.911655000-0400 10.10.1.4 1470 74.53.140.153 25 C: User: Z3VycGFydGFwQHhdHJpb3RzLmlu
2009-10-05T02:06:09.253444000-0400 74.53.140.153 25 10.10.1.4 1470 S: 334 UGFzc3dvcmQ6
2009-10-05T02:06:09.254118000-0400 10.10.1.4 1470 74.53.140.153 25 C: Pass: cHVuamF1QDEyMw==
2009-10-05T02:06:09.613798000-0400 74.53.140.153 25 10.10.1.4 1470 S: 235 Authentication succeeded
2009-10-05T02:06:09.614414000-0400 10.10.1.4 1470 74.53.140.153 25 C: MAIL FROM: <gurpartap@patriots.in>
2009-10-05T02:06:09.956759000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 OK
2009-10-05T02:06:09.957250000-0400 10.10.1.4 1470 74.53.140.153 25 C: RCPT TO: <raj_deol2002in@yahoo.co.in>
2009-10-05T02:06:10.319708000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 Accepted
2009-10-05T02:06:10.320203000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA
2009-10-05T02:06:10.661679000-0400 74.53.140.153 25 10.10.1.4 1470 S: 354 Enter message, ending with "." on a line by itself
2009-10-05T02:06:10.692743000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
2009-10-05T02:06:10.692786000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
2009-10-05T02:06:10.692804000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1460 bytes
2009-10-05T02:06:10.692823000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Window Full] C: DATA fragment, 1460 bytes
2009-10-05T02:06:10.695115000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation needed)
2009-10-05T02:06:10.695623000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation needed)
2009-10-05T02:06:10.696248000-0400 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 Destination unreachable (Fragmentation needed)
2009-10-05T02:06:10.696634000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
2009-10-05T02:06:11.104077000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
2009-10-05T02:06:11.104096000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
2009-10-05T02:06:11.469759000-0400 10.10.1.4 1470 74.53.140.153 25 [TCP Out-Of-Order] C: DATA fragment, 1452 bytes
2009-10-05T02:06:11.469814000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
2009-10-05T02:06:11.494181000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
2009-10-05T02:06:11.494199000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
2009-10-05T02:06:11.834628000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
2009-10-05T02:06:11.834659000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
2009-10-05T02:06:11.858316000-0400 10.10.1.4 1470 74.53.140.153 25 C: DATA fragment, 1452 bytes
2009-10-05T02:06:11.858334000-0400 10.10.1.4 1470 74.53.140.153 25 from: "Gurpartap Singh" <gurpartap@patriots.in>, subject: SMTP, (text/plain) (text/html) (text/plain) (text/plain) | .
2009-10-05T02:06:12.248789000-0400 74.53.140.153 25 10.10.1.4 1470 S: 250 OK id=1Mugho-0003Dg-Un
2009-10-05T02:06:14.763825000-0400 10.10.1.4 1470 74.53.140.153 25 C: QUIT
2009-10-05T02:06:15.105467000-0400 74.53.140.153 25 10.10.1.4 1470 S: 221 xc90.websitewelcome.com closing connection
```

SMTP Conversion Details:

| Item                   | Value                                                    |
|------------------------|----------------------------------------------------------|
| SMTP Server IP         | 74.53.140.153                                            |
| SMTP Server Port       | 25                                                       |
| Client IP              | 10.10.1.4                                                |
| Client Ephemeral Port  | 1470                                                     |
| First SMTP Packet Time | 2009-10-05 02:06:08.219663000 -0400                      |
| Last SMTP Packet Time  | 2009-10-05 02:06:15.105467000 -0400 (closing connection) |
| Number of SMTP Frames  | 27 frames                                                |

Part B: Identify Commands and Response Codes

| <b>Sequence</b> | <b>Direction</b>        | <b>Command/Code</b> | <b>Meaning</b>                | <b>Timestamp</b>                          |
|-----------------|-------------------------|---------------------|-------------------------------|-------------------------------------------|
| <b>1</b>        | <b>Server to client</b> | <b>220</b>          | <b>Service ready</b>          | <b>2009-10-05T02:06:08</b>                |
| <b>2</b>        | <b>Client to server</b> | <b>EHLO/HELO</b>    | <b>Client greeting</b>        | <b>2009-10-05T02:06:08.224809000-0400</b> |
| <b>3</b>        | <b>Client to server</b> | <b>AUTH</b>         | <b>Authentication method</b>  | <b>2009-10-05T02:06:08.568729000-0400</b> |
| <b>4</b>        | <b>Client to server</b> | <b>MAIL FROM</b>    | <b>Envelope sender</b>        | <b>2009-10-05T02:06:09.614414000-0400</b> |
| <b>5</b>        | <b>Client to server</b> | <b>RCPT TO</b>      | <b>Envelope recipient</b>     | <b>2009-10-05T02:06:09.957250000-0400</b> |
| <b>6</b>        | <b>Client to server</b> | <b>DATA</b>         | <b>Message content begins</b> | <b>2009-10-05T02:06:10.320203000-0400</b> |
| <b>7</b>        | <b>Client to server</b> | <b>QUIT</b>         | <b>Session termination</b>    | <b>2009-10-05T02:06:14.763825000-0400</b> |

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab4

(root@kali)~/SBT-DF203-Lab4
# tshark -r working/smtp_working.pcap -Y 'smtp.req || smtp.rsp' -T fields \
-e frame.number -e frame.time -e ip.src -e ip.dst -e smtp.req.command -e smtp.req.parameter \
-e smtp.response.code -e smtp.response.code \
| tee reports/smtp_commands_responses.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
6 2009-10-05T02:06:08.224863000-0400 74.53.140.153 10.10.1.4 220
7 2009-10-05T02:06:08.224863000-0400 10.10.1.4 74.53.140.153 EHLO GP 250
9 2009-10-05T02:06:08.566183000-0400 74.53.140.153 10.10.1.4 AUTH LOGIN 334
10 2009-10-05T02:06:08.568729000-0400 10.10.1.4 74.53.140.153 334
11 2009-10-05T02:06:08.911081000-0400 74.53.140.153 10.10.1.4 334
12 2009-10-05T02:06:08.911655000-0400 10.10.1.4 74.53.140.153 235
13 2009-10-05T02:06:09.253544000-0400 74.53.140.153 10.10.1.4 250
14 2009-10-05T02:06:09.254118000-0400 10.10.1.4 74.53.140.153 MAIL FROM: <gurpartap@patriots.in>
15 2009-10-05T02:06:09.613798000-0400 74.53.140.153 10.10.1.4 250
16 2009-10-05T02:06:09.614414000-0400 10.10.1.4 74.53.140.153 RCPT TO: <raj_deol2002in@yahoo.co.in>
17 2009-10-05T02:06:09.936765000-0400 74.53.140.153 10.10.1.4 250
18 2009-10-05T02:06:09.957250000-0400 10.10.1.4 74.53.140.153 DATA 354
19 2009-10-05T02:06:10.319708000-0400 74.53.140.153 10.10.1.4 250
20 2009-10-05T02:06:10.320203000-0400 10.10.1.4 74.53.140.153 250
21 2009-10-05T02:06:10.661679000-0400 74.53.140.153 10.10.1.4 354
52 2009-10-05T02:06:12.248789000-0400 74.53.140.153 10.10.1.4 250
54 2009-10-05T02:06:14.763825000-0400 10.10.1.4 74.53.140.153 QUIT 221
56 2009-10-05T02:06:15.105467000-0400 74.53.140.153 10.10.1.4 221

(root@kali)~/SBT-DF203-Lab4
```

## Part C - Decode Base64 Authentication Evidence

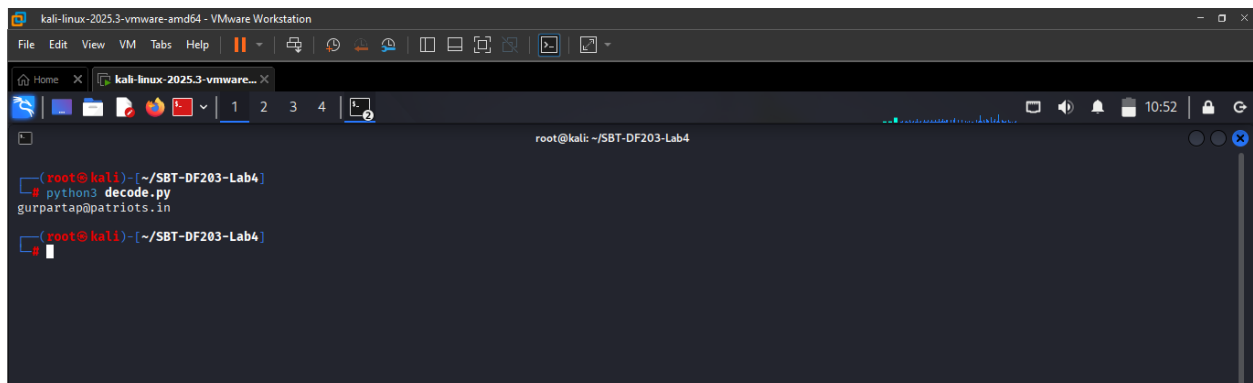
### Using offline decode option

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab4

GNU nano 8.7.1 decode.py
import base64
# base64-encoded text
base64_text = "Z3VycGFydGFwQHBhdHJpb3RzLmlu"
# decode base64-encoded text to ASCII
ascii_text = base64.b64decode(base64_text).decode('utf-8')
# print the decoded ASCII text
print(ascii_text)
```

Decrypted the user base64 code

**Z3VycGFydGFwQHBhdHJpb3RzLmlu**

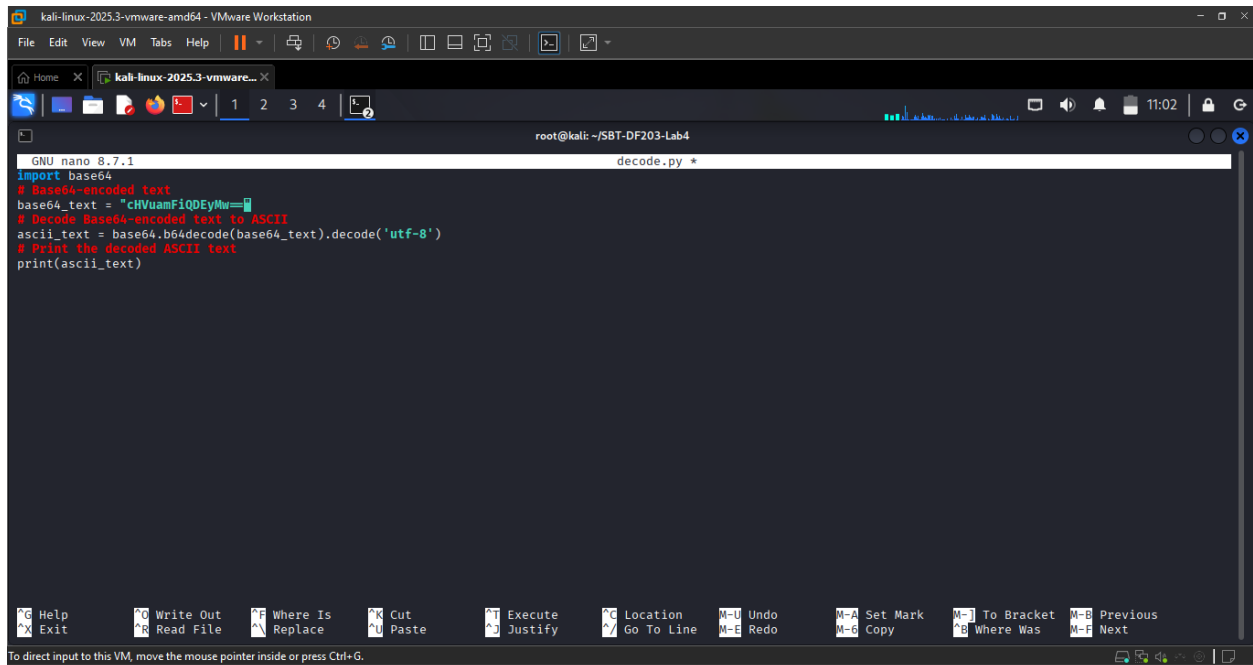


A terminal window titled "kali-linux-2025.3-vmware-amd64 - VMware Workstation" showing a root user at a Kali Linux machine. The user runs a Python script named "decode.py" located in the directory "~/SBT-DF203-Lab4". The script's output is "gurpartap@patriots.in".

```
(root@kali) ~/SBT-DF203-Lab4
# python3 decode.py
gurpartap@patriots.in
(root@kali) ~/SBT-DF203-Lab4
#
```

## Encrypting the Password Base64 code

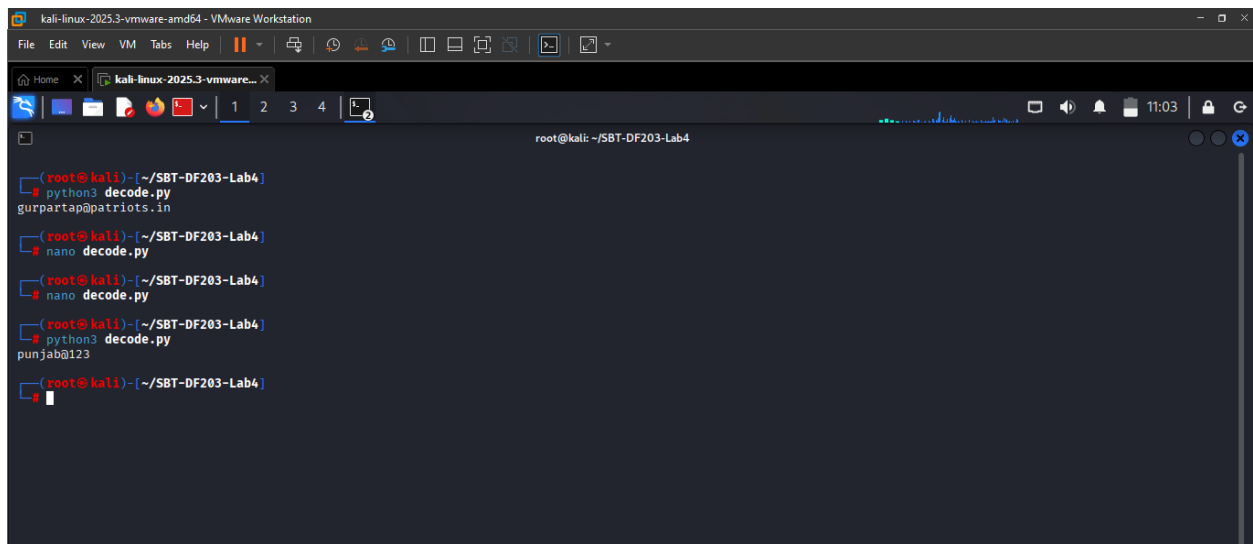
**cHVuamFiQDEyMw==**



A terminal window titled "kali-linux-2025.3-vmware-amd64 - VMware Workstation" showing the content of the "decode.py" script in a nano editor. The script imports the base64 module, defines a Base64-encoded text string, decodes it to ASCII, and prints the result.

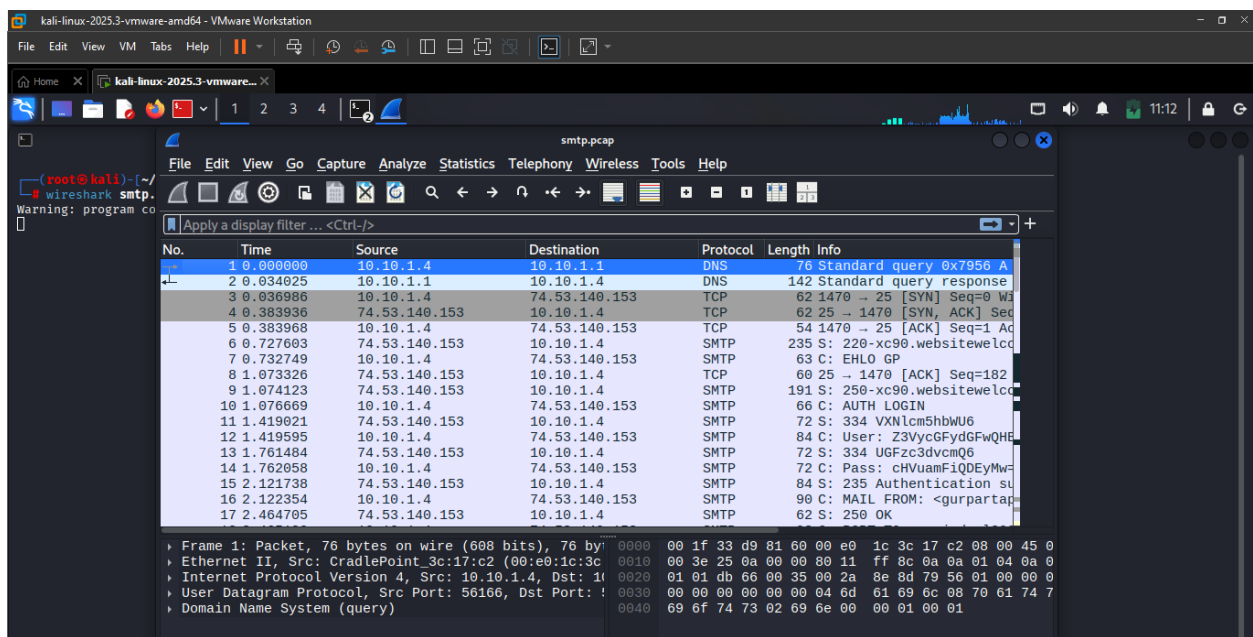
```
GNU nano 8.7.1 decode.py *
import base64
# Base64-encoded text
base64_text = "cHVuamFiQDEyMw=="
# Decode Base64-encoded text to ASCII
ascii_text = base64.b64decode(base64_text).decode('utf-8')
# Print the decoded ASCII text
print(ascii_text)
```

At the bottom of the terminal, there is a list of keyboard shortcuts for the nano editor, such as "H Help", "X Exit", "O Write Out", "R Read File", "W Where Is", "A Replace", "C Cut", "U Paste", "E Execute", "J Justify", "G Location", "o Go To Line", "U Undo", "E Redo", "A Set Mark", "C Copy", "I To Bracket", "B Where Was", "P Previous", "F Next".

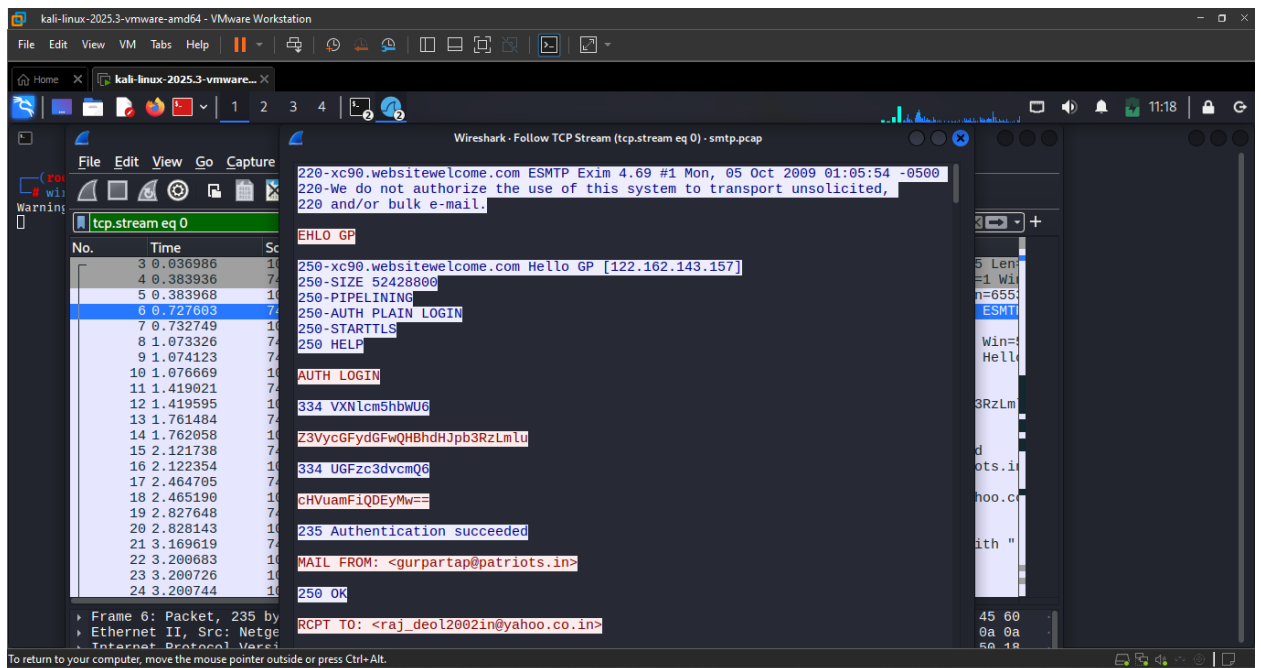


## Part D - Reconstructing Email Message

### 1. Capturing the smtp file



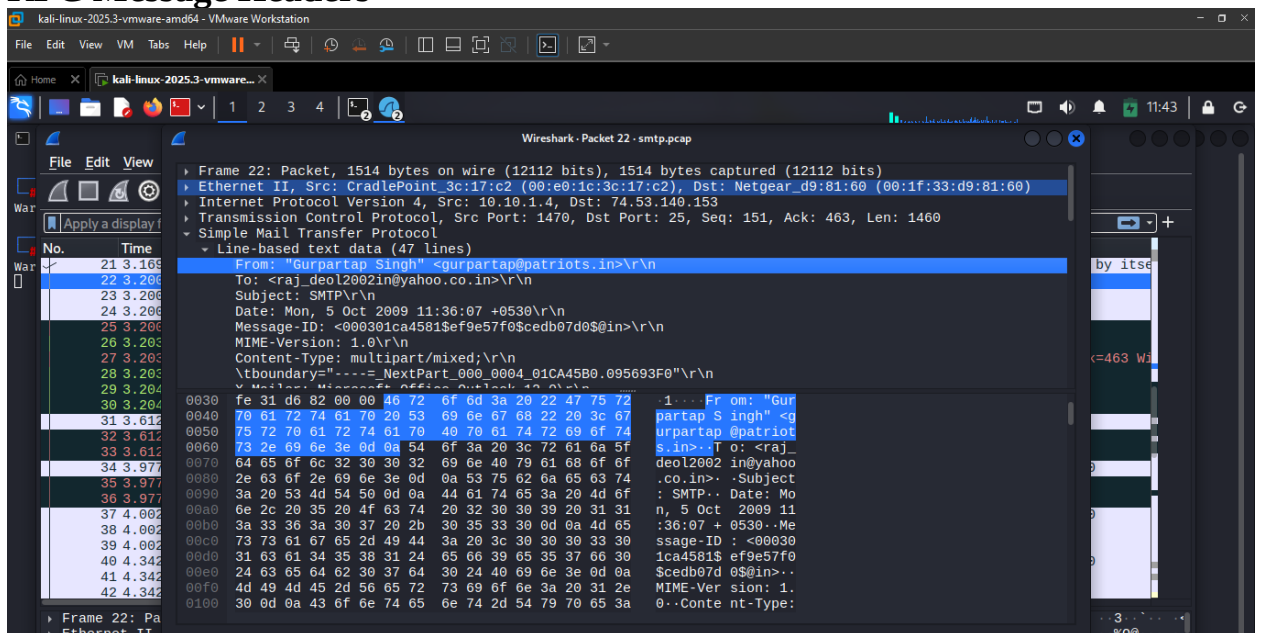
### 2. selecting an SMTP packet and choosing to Follow > TCP Stream.



### 3. identifying SMTP envelope fields

| Envelope Field | Value                        |
|----------------|------------------------------|
| MAIL FROM      | <gurpartap@patriots.in>      |
| RCPT TO        | <raj_deol2002in@yahoo.co.in> |

### 4. RFC Message Headers



| Header            | Value                                            |
|-------------------|--------------------------------------------------|
| Date              | Mon, 5 Oct 2009 11:36:07 +0530\r\n               |
| From              | "Gurpartap Singh"<br><gurpartap@patriots.in>\r\n |
| To                | <raj_deol2002in@yahoo.co.in>\r\n                 |
| Subject           | SMTP\r\n                                         |
| Message ID        | <000301ca4581\$ef9e57f0\$cedb07d0\$@in>\r\n      |
| MIME Version      | 1.0\r\n                                          |
| Content Type      | multipart/mixed;\r\n                             |
| User Agent/mailer | Microsoft Office Outlook 12.0\r\n                |
| Content-Language  | en-us\r\n                                        |

```

> Internet Protocol Version 4, Src: 10.10.1.4, Dst: 74.53.140.153
> Transmission Control Protocol, Src Port: 1470, Dst Port: 25, Seq: 151, Ack: 463, Len: 1460
> Simple Mail Transfer Protocol
  > Line-based text data (47 lines)
    From: "Gurpartap Singh" <gurpartap@patriots.in>\r\n
    To: <raj_deol2002in@yahoo.co.in>\r\n
    Subject: SMTP\r\n
    Date: Mon, 5 Oct 2009 11:36:07 +0530\r\n
    Message-ID: <000301ca4581$ef9e57f0$cedb07d0$@in>\r\n
    MIME-Version: 1.0\r\n
    Content-Type: multipart/mixed;\r\n
    \tboundary="-----_NextPart_000_0004_01CA45B0.095693F0"\r\n
    X-Mailer: Microsoft Office Outlook 12.0\r\n
    Thread-Index: AcpFgem9BvjZEDeR1Kh8i+hUyVo0A==\r\n
    Content-Language: en-us\r\n
    [...] x-cr-hashedpuzzle: SeA= AAR2 ADaH Bpi0 C4G1 D1gW FNB1 FPKR Fn+W HFcp HnYJ J07s Kum6 KytW LFcI LjU
    x-cr-puzzleid: {CAA37F59-1850-45C7-8540-AA27696B5398}\r\n
    \r\n
    This is a multipart message in MIME format.\r\n
    \r\n
    -----_NextPart_000_0004_01CA45B0.095693F0\r\n
    Content-Type: multipart/alternative;\r\n
    ...
01b0 3d 0d 0a 43 6f 6e 74 65 6e 74 2d 4c 61 6e 67 75 =...Conte nt-Langu
01c0 61 67 65 3a 20 65 6e 2d 75 73 0d 0a 78 2d 63 72 age: en- us- x-cr
01d0 2d 68 61 73 68 65 64 70 75 7a 7a 6c 65 3a 20 53 -hashedp uzzle: S
01e0 65 41 3d 20 41 41 52 32 20 41 44 61 48 20 42 70 eA= AAR2 ADaH Bp
01f0 69 4f 20 43 34 47 31 20 44 31 67 57 20 46 4e 42 i0 C4G1 D1gW FNB
0200 31 20 46 50 6b 52 20 46 6e 2b 57 20 48 46 43 50 1 FPKR F n+W HFcp
Text item (body): 25 bytes

```

## Part E: Determine Client, Hosts and Network Metadata

```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab4

(root@kali) ~/SBT-DF203-Lab4
# tshark -r working/smtp_working.pcap -Y 'smtp' -T fields \
-e frame.number -e frame.time_epoch -e eth.src -e eth.dst -e ip.src -e tcp.srcport \
-e ip.dst -e tcp.dstport -e tcp.stream \
| tee reports/smtp_network_metadata.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
6 1254722768.219663000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
7 1254722768.224809000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
9 1254722768.566183000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
10 1254722768.568729000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
11 1254722768.911081000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
12 1254722768.911655000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
13 1254722769.253544000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
14 1254722769.254118000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
15 1254722769.613798000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
16 1254722769.614414000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
17 1254722769.956765000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
18 1254722769.957250000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
19 1254722770.319708000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
20 1254722770.320203000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
21 1254722770.661679000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
22 1254722770.692743000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
23 1254722770.692786000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
24 1254722770.692804000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
25 1254722770.692823000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
26 1254722770.695115000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
28 1254722770.695623000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
29 1254722770.696248000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
30 1254722770.696634000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
32 1254722771.104972000 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
33 1254722771.104998000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
```

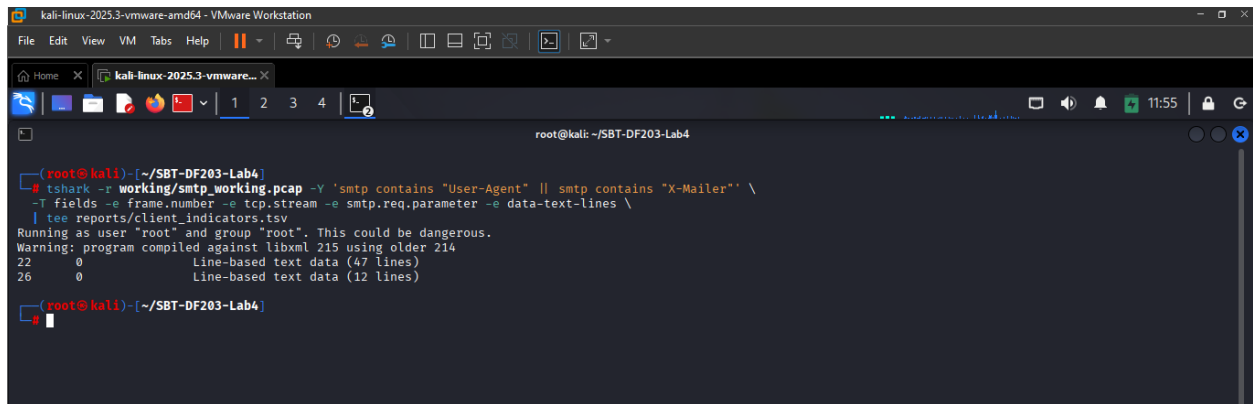
```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
1 2 3 4
root@kali: ~/SBT-DF203-Lab4

13 1254722769.253544000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
14 1254722769.254118000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
15 1254722769.613798000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
16 1254722769.614414000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
17 1254722769.956765000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
18 1254722769.957250000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
19 1254722770.319708000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
20 1254722770.320203000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
21 1254722770.661679000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
22 1254722770.692743000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
23 1254722770.692786000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
24 1254722770.692804000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
25 1254722770.692823000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
26 1254722770.695115000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
28 1254722770.695623000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
29 1254722770.696248000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
30 1254722770.696634000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 192.168.1.1,10.10.1.4 1470 10.10.1.4,74.53.140.153 25 0
32 1254722771.104972000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
33 1254722771.104998000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
35 1254722771.469759000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
36 1254722771.469814000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
38 1254722771.494181000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
39 1254722771.494199000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
41 1254722771.834628000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
42 1254722771.834655000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
44 1254722771.858316000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
45 1254722771.858334000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
52 1254722772.248789000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0
54 1254722774.763825000 00:e0:1c:3c:17:c2 00:1f:33:d9:81:60 10.10.1.4 1470 74.53.140.153 25 0
56 1254722775.105467000 00:1f:33:d9:81:60 00:e0:1c:3c:17:c2 74.53.140.153 25 10.10.1.4 1470 0

(root@kali) ~/SBT-DF203-Lab4
```

## Part F - Encryption and Evidential Limitations

STARTTLS is requested and subsequent payload becomes TLS.



```
kali-linux-2025.3-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
kali-linux-2025.3-vmware...
root@kali: ~/SBT-DF203-Lab4
(root@kali)~/SBT-DF203-Lab4
# tshark -r working/smtp_working.pcap -Y 'smtp contains "User-Agent" || smtp contains "X-Mailer"' \
-T fields -e frame.number -e tcp.stream -e smtp.req.parameter -e data-text-lines \
| tee reports/client_indicators.tsv
Running as user "root" and group "root". This could be dangerous.
Warning: program compiled against libxml 215 using older 214
22      0      Line-based text data (47 lines)
26      0      Line-based text data (12 lines)
(root@kali)~/SBT-DF203-Lab4
#
```

**Analysis of the SMTP traffic showed no STARTTLS negotiation. The session proceeded directly from EHLO to AUTH LOGIN and subsequent SMTP commands without a STARTTLS request. No TLS handshake packets were observed, and the protocol remained SMTP throughout the capture. Therefore, the connection was not upgraded to TLS, and the SMTP payload was transmitted in plaintext.**